Travel agency · flights from New York

Can we tell which flights will go wrong, two hours before they leave?

What we found in the 2023 flights from JFK, LGA and EWR, what we can predict, and what the agency can do about it.

435,352

flights analysed

Nearly 1 flight in 4 leaves late or is cancelled

22.9%

of flights left more than 15 minutes late, or never left

If we knew two hours before which flights are at risk, we could warn the client and arrange a backup.

Late means

More than 15 minutes after the scheduled time, or the flight never leaves.

Why it matters

A client misses a connection or a meeting.

Our two questions

Will this flight be late or cancelled? And how many minutes late will it be?

We checked the data before trusting it, and lost no flight

14 checks

run on the data, with a decision for each problem

All 435,352 flights were kept.

10,738 cancelled flights

Kept and marked. They are exactly the flights we want to warn about.

1,913 flights with no plane number, 826 with no weather reading

Nothing was thrown away: the gaps are marked and handled.

What the data cannot tell us

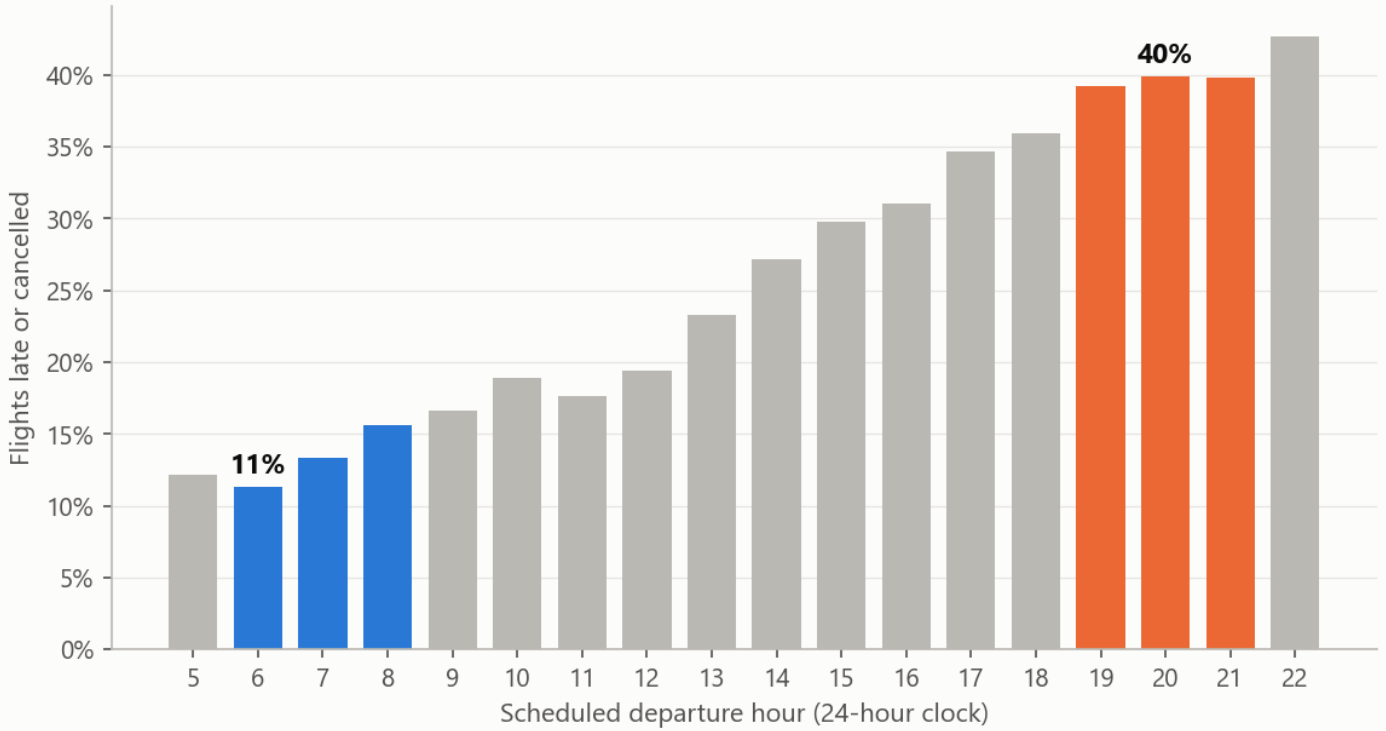
Only one year, only flights leaving New York, no passengers or prices, no reasons for delays.

Delays build up through the day

13% → 40%

late or cancelled: flights at 6-9 am against 7-10 pm

Flights scheduled in the morning are rarely late; in the evening far more of them are. **Clients with tight connections should fly in the morning.**

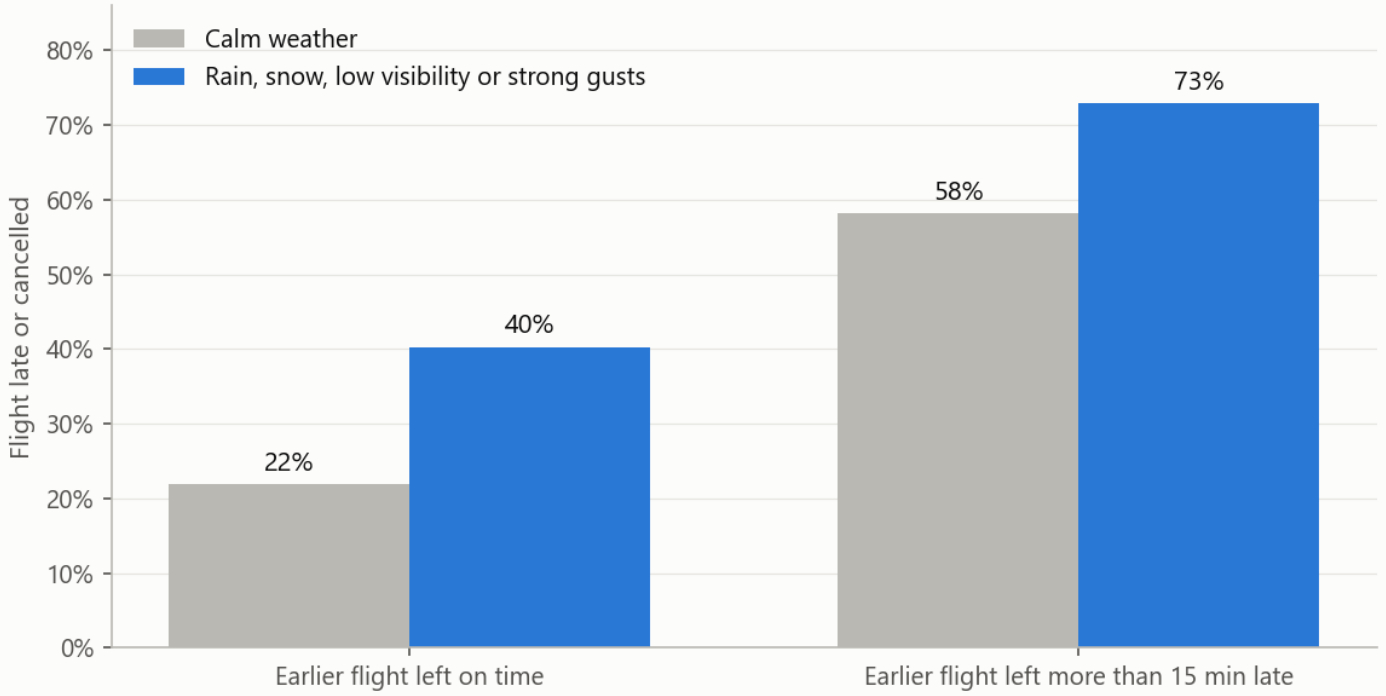


A late earlier flight is a strong warning sign, stronger than bad weather

+36 vs +18

points of extra risk, each measured with the other absent: a late earlier flight against bad weather

When the same plane's earlier flight left more than 15 minutes late, **61%** of its next flights are late or cancelled, against **25%** otherwise. Part of that is a shared bad day: within the same day, airport and part of the day, the overall gap of 37 points shrinks to 26. The agency can see the earlier flight two hours ahead.

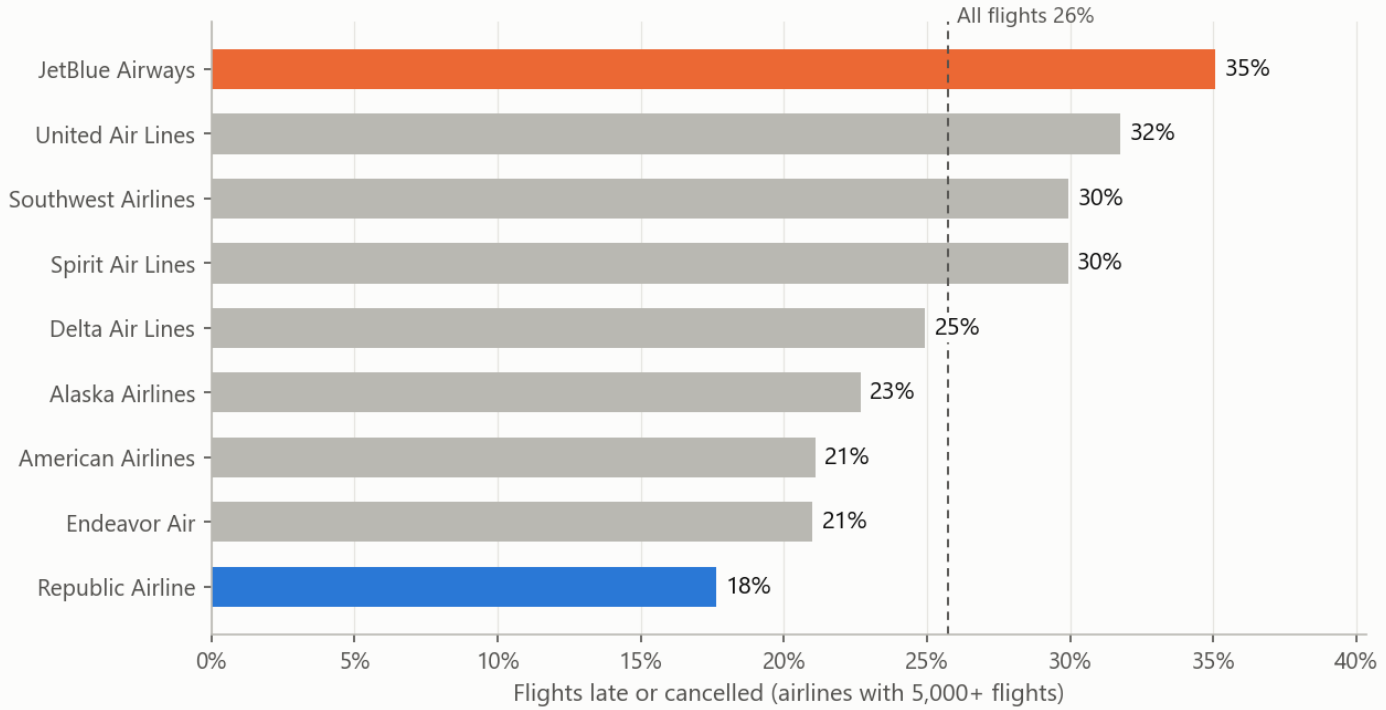


Airlines differ a lot, and airport and time of day do not explain it

35% vs 18%

late or cancelled: JetBlue Airways against Republic Airline

Even when we compare flights from the same airport at the same time of day, the gap remains. **For a tight connection, prefer the steadier airlines.**



Destinations fall into three types, each needing a different action

64 steady routes

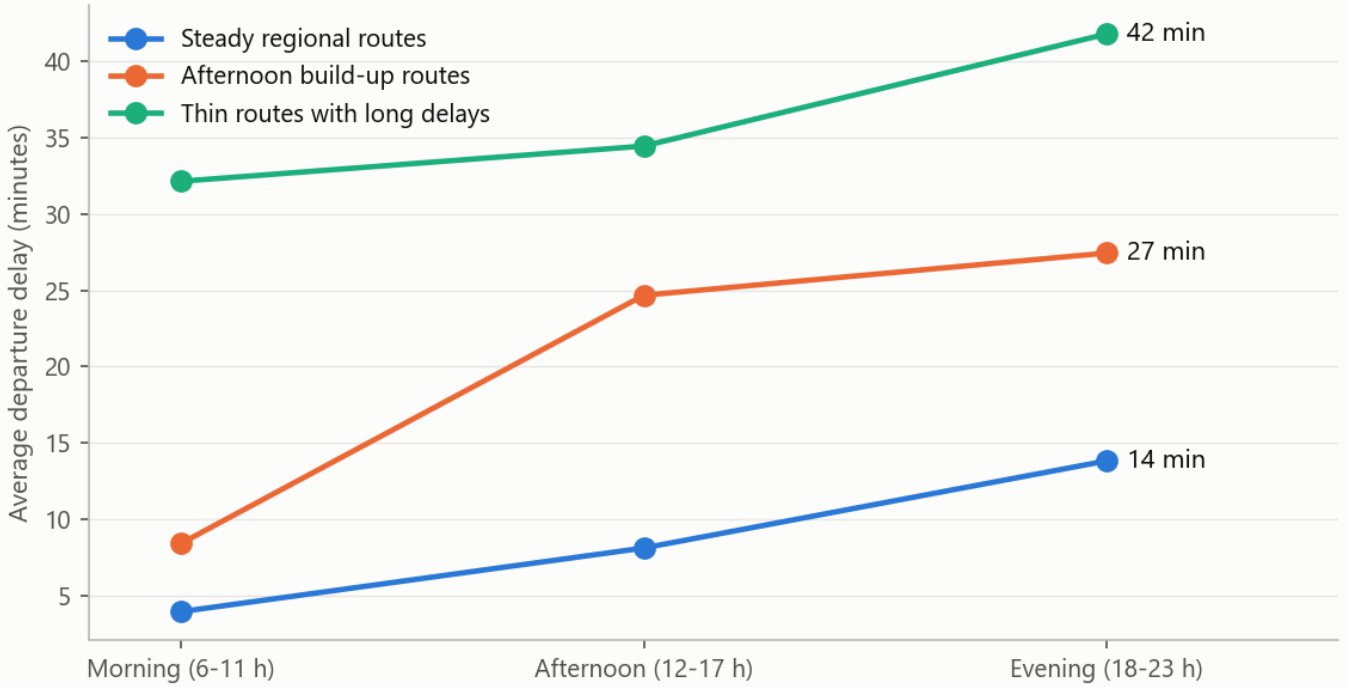
Short delays all day, but a few more cancellations. Keep a backup for tight connections.

39 afternoon build-up routes

Fine in the morning, late after noon. Offer a morning flight.

6 thin routes

About one flight a day, 53% of them late. Offer a flexible ticket.



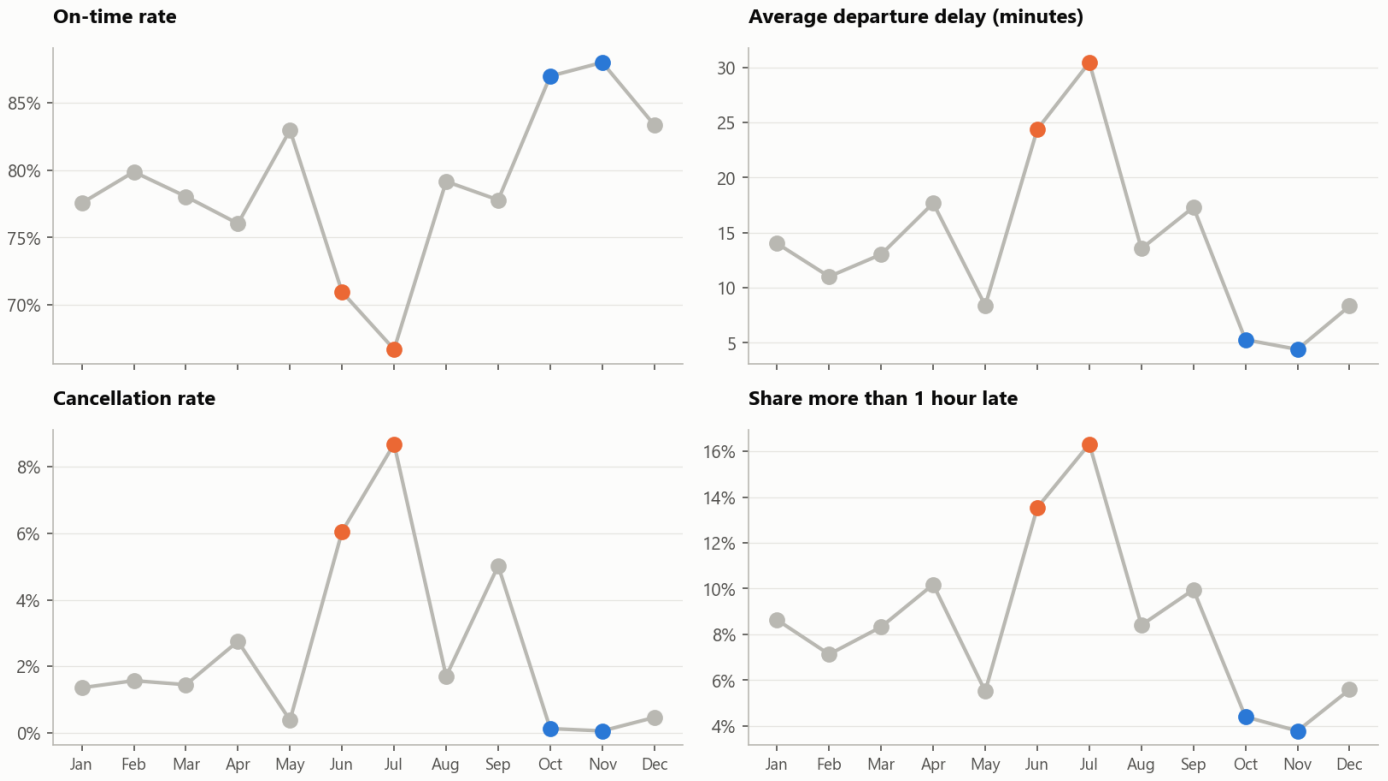
The groups are a simplification: they shift when built from other months, and 'thin routes' is the least stable.

July was the worst month and autumn the calmest

39% vs 12%

late or cancelled: July against November. These are the monthly numbers to follow.

In July **8.7%** of flights were cancelled. Just 10 days hold **35%** of all cancellations up to September, and the weather at the airports does not fully explain them: the cause is not in our data.



Prediction 1: warn 2 flights in 10 and reach 2.0 times as many late ones as luck

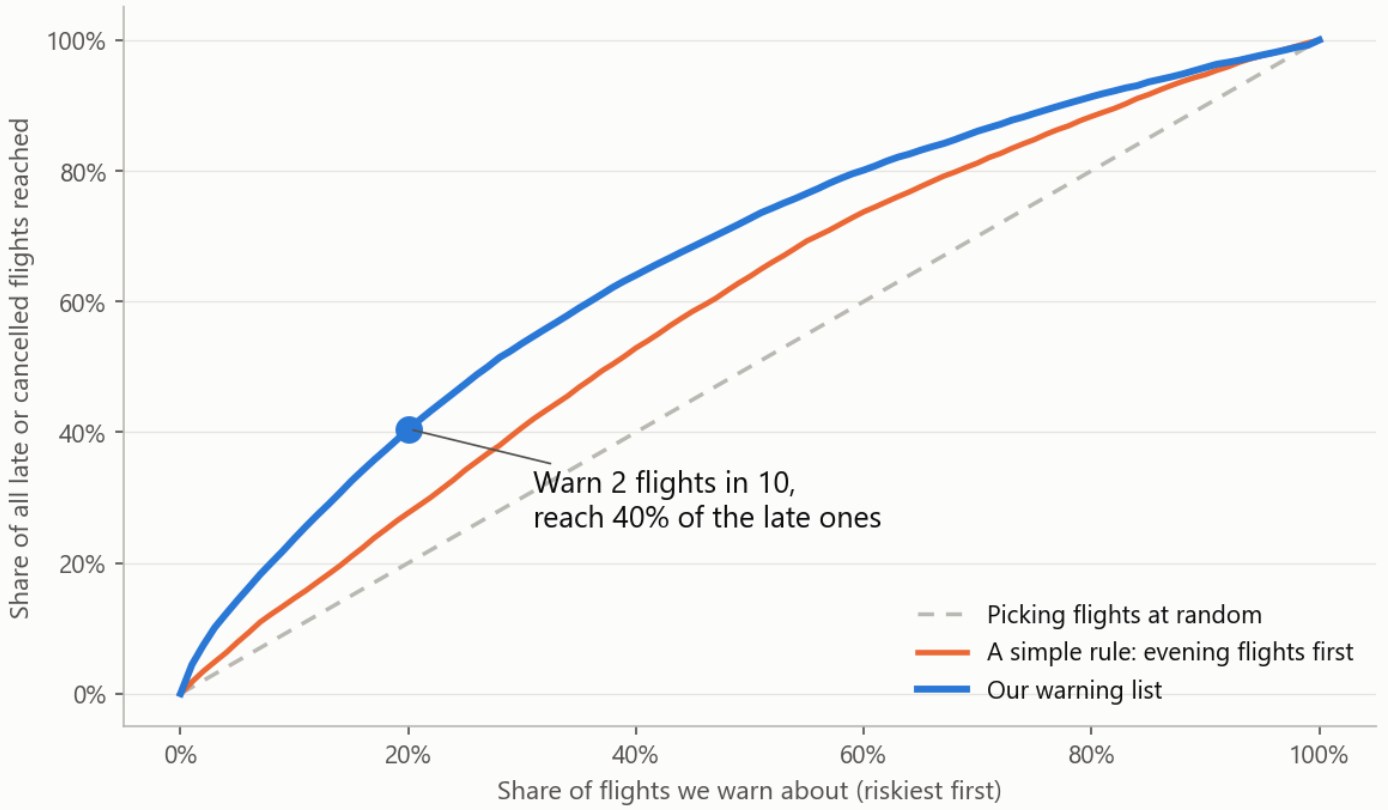
40% vs 20%

of the late or cancelled flights reached, against picking at random

In October to December we would have warned **20,894** flights (2 in 10) and reached **5,947** late or cancelled ones, against about **2,937** by luck. 28 warnings in 100 were right; 14 flights in 100 were late. Run live, with the cut-off set from the previous 7 days, it reaches **40%**.

Learn and tune
Jan-Sep

Test, kept aside
Oct-Dec

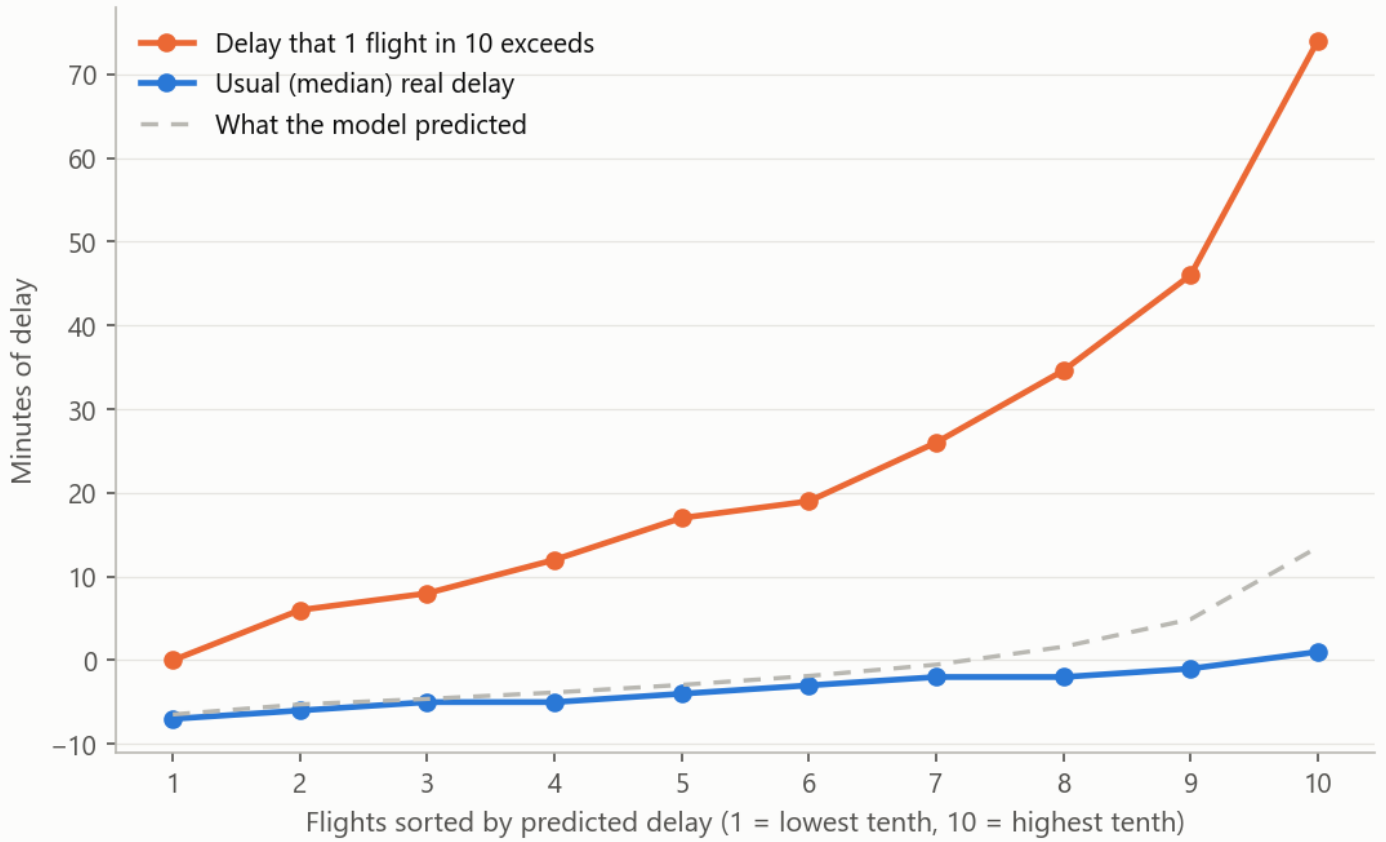


Prediction 2: how late will a flight be? A modest gain, honestly reported

3.4×

more long delays (45+ minutes) among the 5% of flights with the highest predicted delay: 21% against 6%

On average the predicted delay is only **0.14 minutes** closer than a simple guess, because a few extreme delays cannot be foreseen. **We use it to rank flights for a connection check, not to promise a number of minutes.**



Two decision cards: what to do, for how many flights, and when to re-check

Card 1 · warn the client

Who: the 2 in 10 riskiest flights, 20,894 in 3 months.

Do: warn the client and check the connection. A person confirms first.

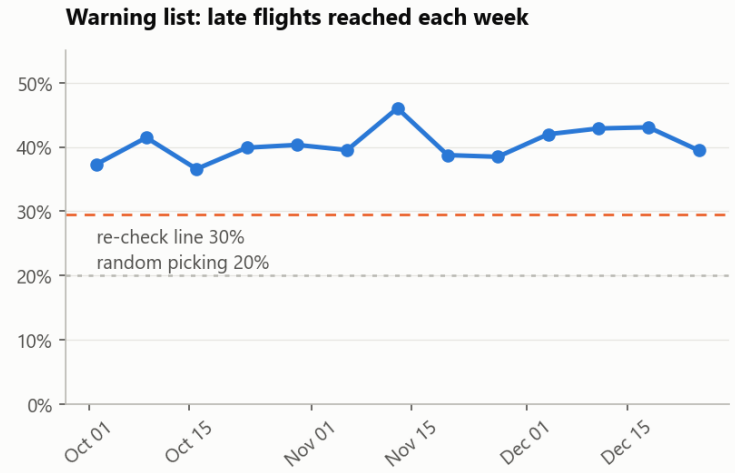
Re-check if the weekly reach falls below **30%** two weeks in a row. It stayed between 37% and 46%.

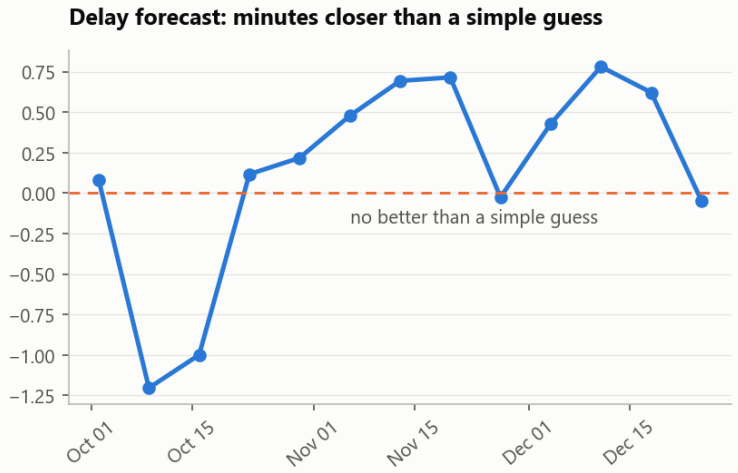
Card 2 · check the connection

Who: the 1 in 20 flights with the highest predicted delay, 5,212 in 3 months.

Do: tell the client to expect a long delay and prepare a backup.

Re-check or drop it if it is no better than a simple guess two weeks in a row: its gain comes and goes.





Assumptions to confirm with the agency: it can warn about 2 flights in 10 and check 1 in 20. If not, we change one number and everything updates.

What we do not know, what comes next, and how AI helped

What we do not know

- The test months were calm: the model expected **22%** late flights, **14%** were. We trust the ranking, not the percentages.
- It reaches only **7%** of the late morning flights, but **70%** of the evening ones.
- One year of data, no prices, no reasons for delays.

What comes next

- Confirm the two capacities with the agency.
- Add a second year, so seasons can be learned.
- Set a separate warning quota for each part of the day.
- Add the agency's own costs to put a value on each warning.

How AI helped

Claude Code, an AI assistant, wrote the code, ran the analysis and drafted these slides, following my instructions. It also recorded its own mistakes and corrections.

What I checked myself, on an earlier version of the project:

- re-ran everything from scratch; the checks passed
- recomputed a headline number from the raw data
- recomputed the value of the warning list
- read how the models are kept from using future information
- confirmed the test months stay untouched until the end
- read the leak found by the independent review and how it was fixed
- went through all 8 notebooks and the log of mistakes

The project changed after that; I have not yet repeated these checks on the final version.